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The Age Dependence of the Spiral J_R–h_Z Relation

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ABSTRACT

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1. INTRODUCTION
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3. METHOD
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We use the density fits of J. Imig et al. (2025), released under CC BY 4.0 at https://github.com/astrojimig/mw_density_imig2025.

REFERENCES

Abdurro'uf, Accetta, K., Aerts, C., et al. 2022, ApJS, 259, 35, doi: 10.3847/1538-4365/ac4414

Aumer, M., Binney, J., & Schönrich, R. 2016, MNRAS, 462, 1697

Binney, J. 2012, MNRAS, 426, 1324, doi: 10.1111/j.1365-2966.2012.21757.x

Binney, J., & Tremaine, S. 2008, Galactic Dynamics: Second Edition (Princeton Univ. Press)

Blanton, M. R., Bershady, M. A., Abolfathi, B., et al. 2017, AJ, 154, 28, doi: 10.3847/1538-3881/aa7567

Fattahi, A., Belokurov, V., Deason, A. J., et al. 2019, MNRAS, 484, 4471, doi: 10.1093/mnras/stz159

Gaia Collaboration, Prusti, T., de Bruijne, J. H. J., et al. 2016, A&A, 595, A1, doi: 10.1051/0004-6361/201629272

Gaia Collaboration, Vallenari, A., Brown, A. G. A., et al. 2023, A&A, 674, A1, doi: 10.1051/0004-6361/202243940

Grand, R. J. J., Fragkoudi, F., Gómez, F. A., et al. 2024, MNRAS, 532, 1814, doi: 10.1093/mnras/stae1598

Grand, R. J. J., Gómez, F. A., Marinacci, F., et al. 2017, MNRAS, 467, 179, doi: 10.1093/mnras/stx071

Gunn, J. E., Siegmund, W. A., Mannery, E. J., et al. 2006, AJ, 131, 2332, doi: 10.1086/500975

Huang, Y., Schönrich, R., Zhang, H., et al. 2020, ApJS, 249, 29, doi: 10.3847/1538-4365/ab994f

Imig, J., Holtzman, J. A., Zasowski, G., Lian, J., et al. 2025, ApJ, 990, 203, doi: 10.3847/1538-4357/adf723

Jia, Y., Yang, C., Chen, Y., Du, C., & Zhao, G. 2024, A&A, 692, A167

Leung, H. W., & Bovy, J. 2019a, MNRAS, 483, 3255, doi: 10.1093/mnras/sty3217

Leung, H. W., & Bovy, J. 2019b, MNRAS, 489, 2079, doi: 10.1093/mnras/stz2245

Ludlow, A. D., Fall, S. M., Schaye, J., & Obreschkow, D. 2021, MNRAS, 508, 5114, doi: 10.1093/mnras/stab2770

Mackereth, J. T., Bovy, J., Leung, H. W., et al. 2019, MNRAS, 489, 176, doi: 10.1093/mnras/stz1521

Majewski, S. R., Schiavon, R. P., Frinchaboy, P. M., et al. 2017, AJ, 154, 94, doi: 10.3847/1538-3881/aa784d

Palicio, P. A., Recio-Blanco, A., Tepper-García, T., et al. 2025, A&A, 695, A193

Skrutskie, M. F., Cutri, R. M., Stiening, R., et al. 2006, AJ, 131, 1163, doi: 10.1086/498708

Spitzer, L. 1942, ApJ, 95, 329

Stone-Martinez, A., Holtzman, J. A., Imig, J., et al. 2024, AJ, 167, 73, doi: 10.3847/1538-3881/ad12a6

Sun, W., Shen, H., Jiang, B., & Liu, X. 2025, ApJ, 979, 103

Vasiliev, E. 2019, MNRAS, 482, 1525, doi: 10.1093/mnras/sty2672

Vera-Ciro, C., D'Onghia, E., & Navarro, J. F. 2016, ApJ, 833, 42

Wilson, J. C., Hearty, F. R., Skrutskie, M. F., et al. 2019, PASP, 131, 055001, doi: 10.1088/1538-3873/ab0075